

UNIVERSIDAD DE SAN CARLOS DE GUATEMALA

Facultad de Ingeniería

Escuela de Ingeniería en Ciencias y Sistemas

0970 - Redes de Computadoras 1

Laboratorio - Segundo Semestre 2026

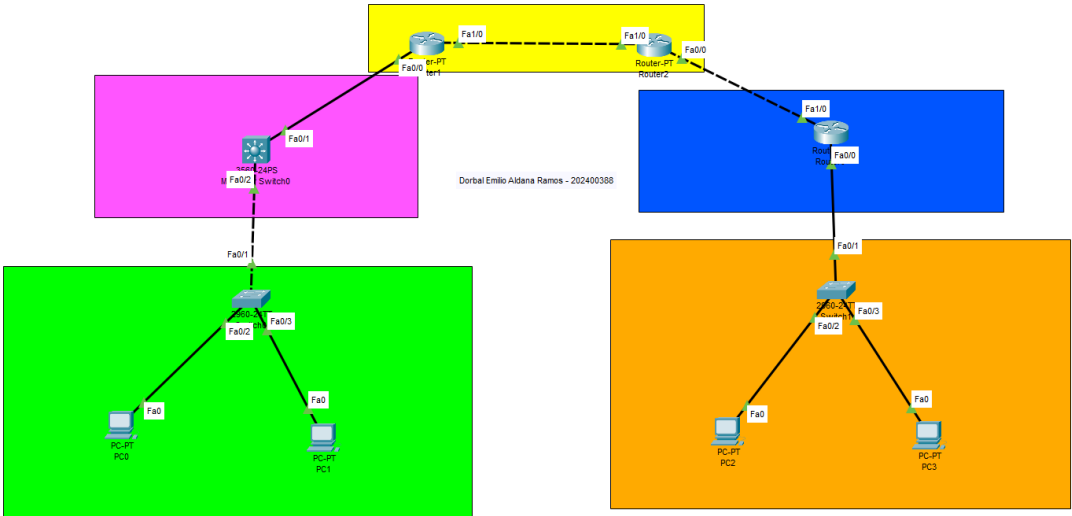
TAREA #4

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Topología final



Configuración y verificación

Switch del área verde

Diagram illustrating the configuration and verification of the switch in the green area. The switch is connected to a router and two PCs. The CLI window shows the configuration of VLANs and interfaces.

```
Switch#show vlan brief
```

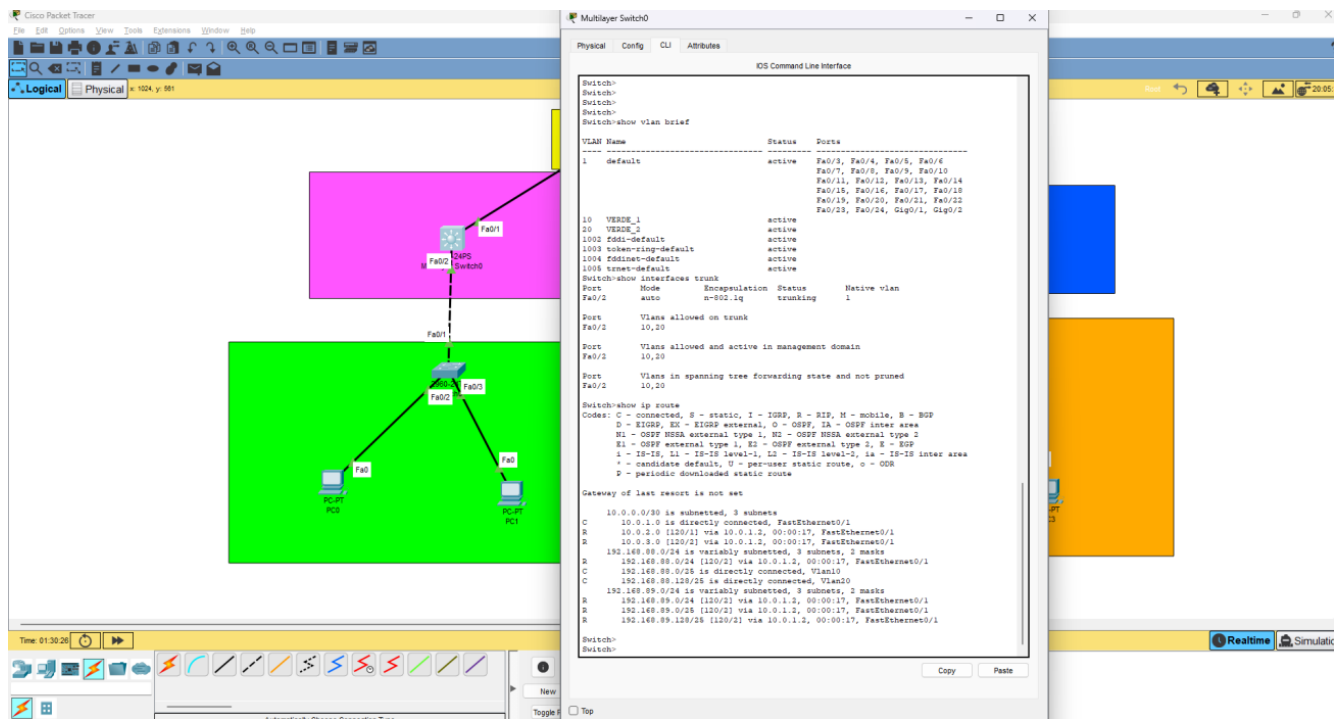
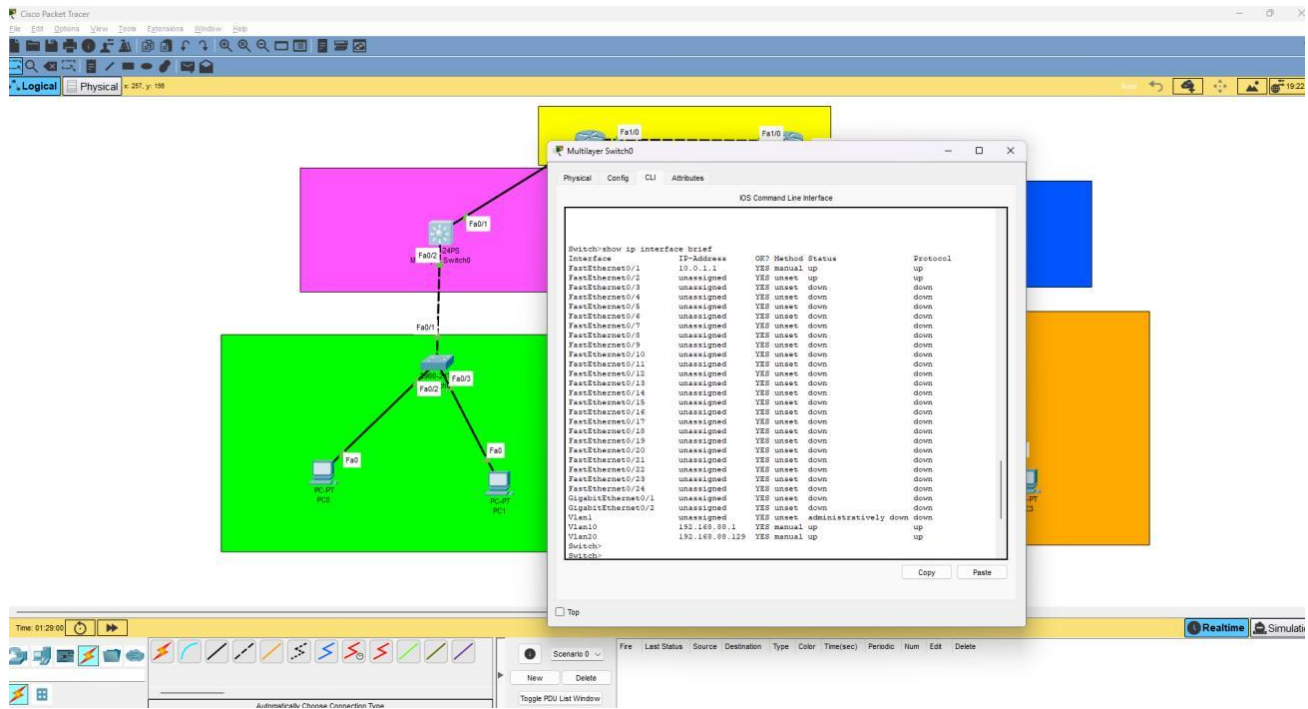
VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 VESDE_1	active	Fa0/2
20 VESDE_2	active	Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 token-ring-default	active	

```
Switch#show interface trunk
```

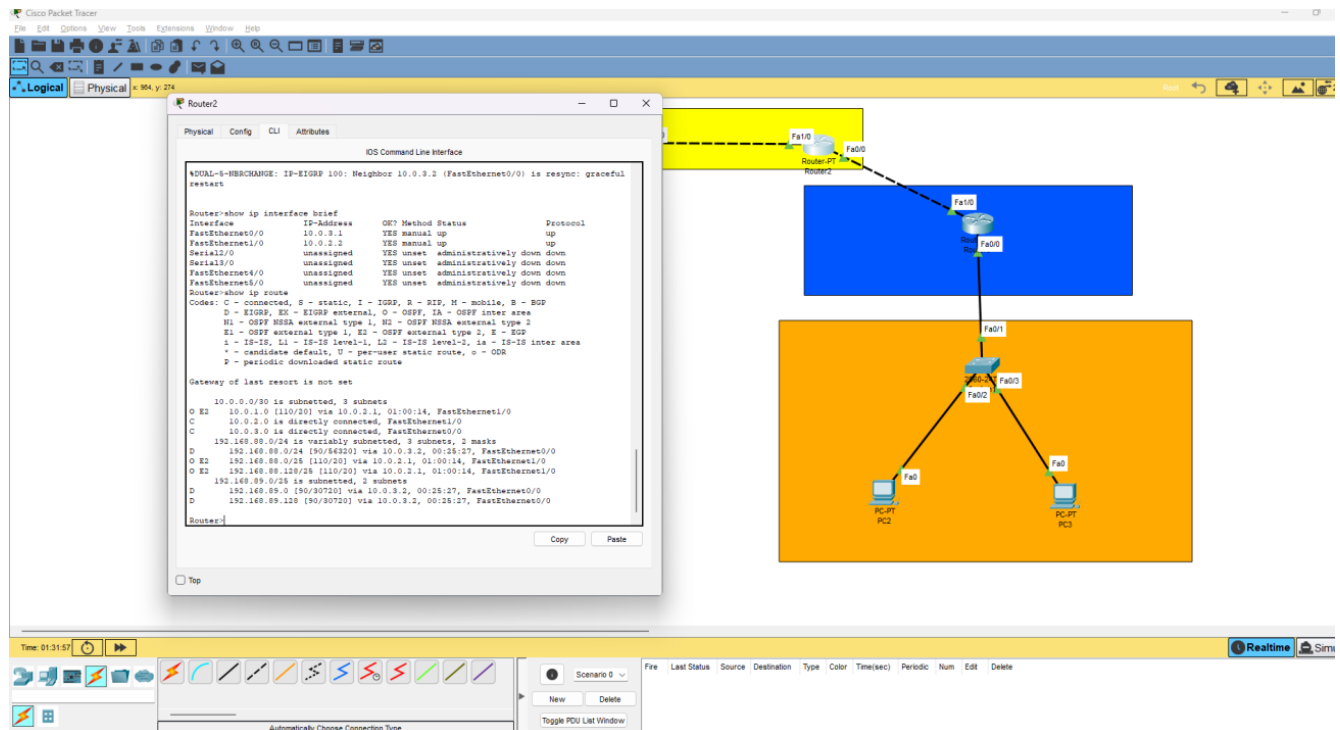
Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

```
Switch#
```

Switch del area morada



Router1



Router0 - área azul

Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   up           up
FastEthernet0/30 192.168.89.1    YES manual  up           up
FastEthernet0/40 192.168.89.129 YES manual  up           up
FastEthernet1/0 10.0.3.2        YES manual  up           up
Serial1/0        unassigned      YES unset   administrati down
FastEthernet4/0 unassigned      YES unset   administrati down
FastEthernet5/0 unassigned      YES unset   administrati down
FastEthernet6/0 unassigned      YES unset   administrati down

Router>show ip route
Codes: C - connected, S - static, I - IGMP, D - RIP, H - mobile, B - BGP
        O - OSPF, EX - OSPF external, D - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EIGRP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        S - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 3 subnets
D EX 10.0.1.0 [170/83760] via 10.0.3.1, 00:24:02, FastEthernet1/0
D EX 10.0.2.0 [170/83760] via 10.0.3.1, 00:24:02, FastEthernet1/0
C      10.0.3.0 is directly connected, FastEthernet1/0
192.168.89.0/24 is variably subnetted, 3 subnets, 2 masks
D      192.168.89.0/24 is a summary, 00:24:11, Null0
D EX 192.168.89.0/28 [170/83760] via 10.0.3.1, 00:24:02, FastEthernet1/0
D EX 192.168.89.128/28 [170/83760] via 10.0.3.1, 00:24:02, FastEthernet1/0
192.168.89.0/25 is subnetted, 2 subnets
D EX 192.168.89.0 is directly connected, FastEthernet0/0,30
C      192.168.89.128 is directly connected, FastEthernet0/40

Router>
```

Switch del área naranja

Switch1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Switch>show vlan brief
VLAN Name         Status Ports
-----
1  default       active Fa0/4, Fa0/6, Fa0/7,
                        Fa0/9, Fa0/9, Fa0/10, Fa0/11,
                        Fa0/12, Fa0/13, Fa0/14, Fa0/16,
                        Fa0/16, Fa0/17, Fa0/18, Fa0/19,
                        Fa0/10, Fa0/21, Fa0/22, Fa0/23,
                        Fa0/24, Gig0/1, Gig0/2
30  NARANJA_1     active Fa0/3
40  NARANJA_2     active Fa0/3
1003 cddi-default active
1004 token-ring-default active
1005 tynet-default active

Switch>show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     30,40

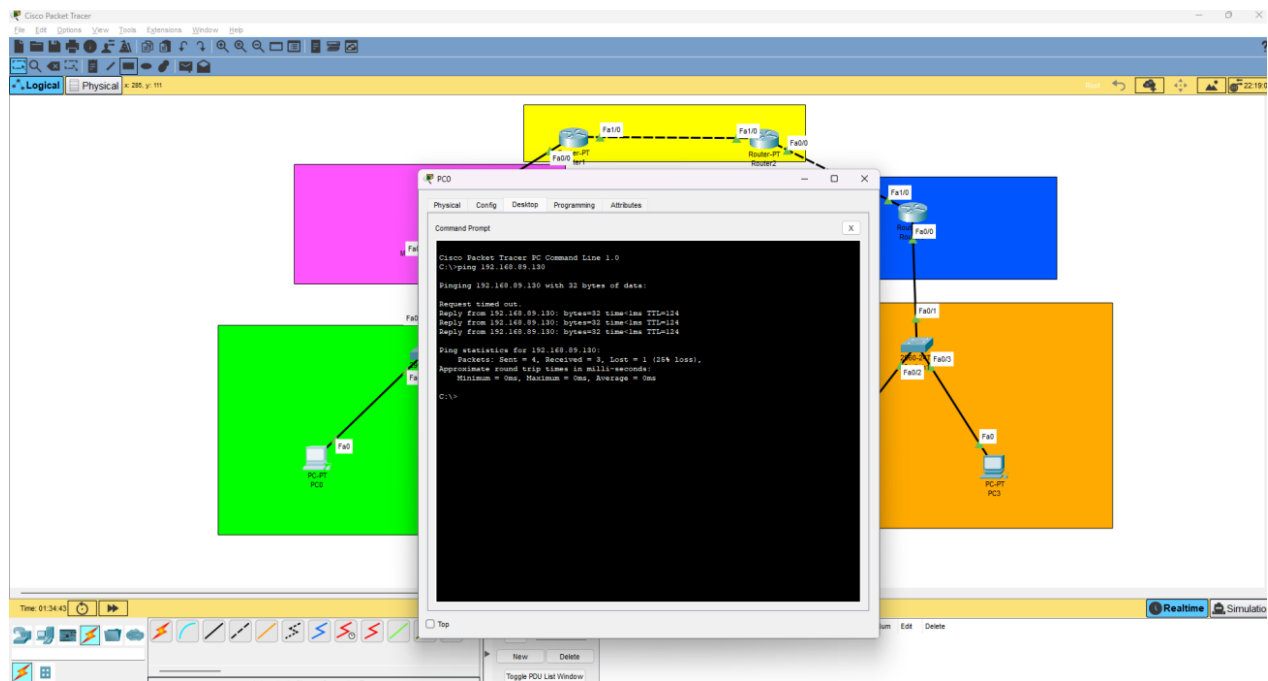
Port      Vlans allowed and active in management domain
Fa0/1     30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     30,40

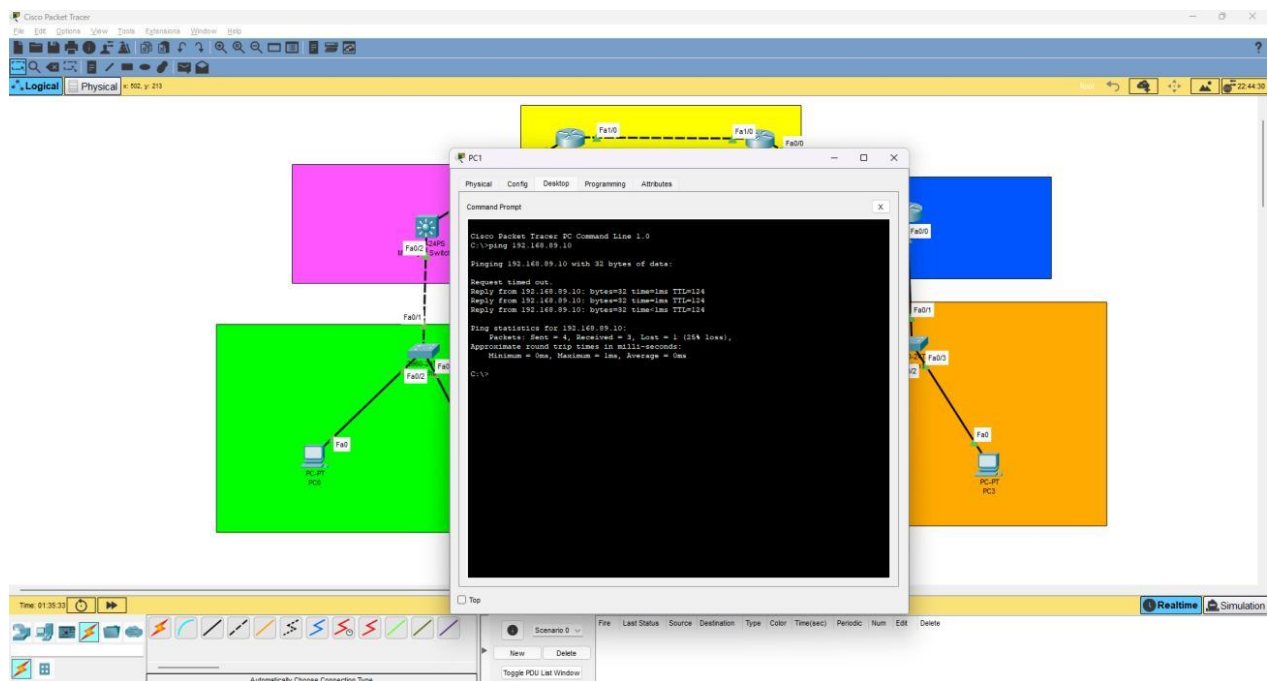
Switch>
```

Pruebas de conectividad

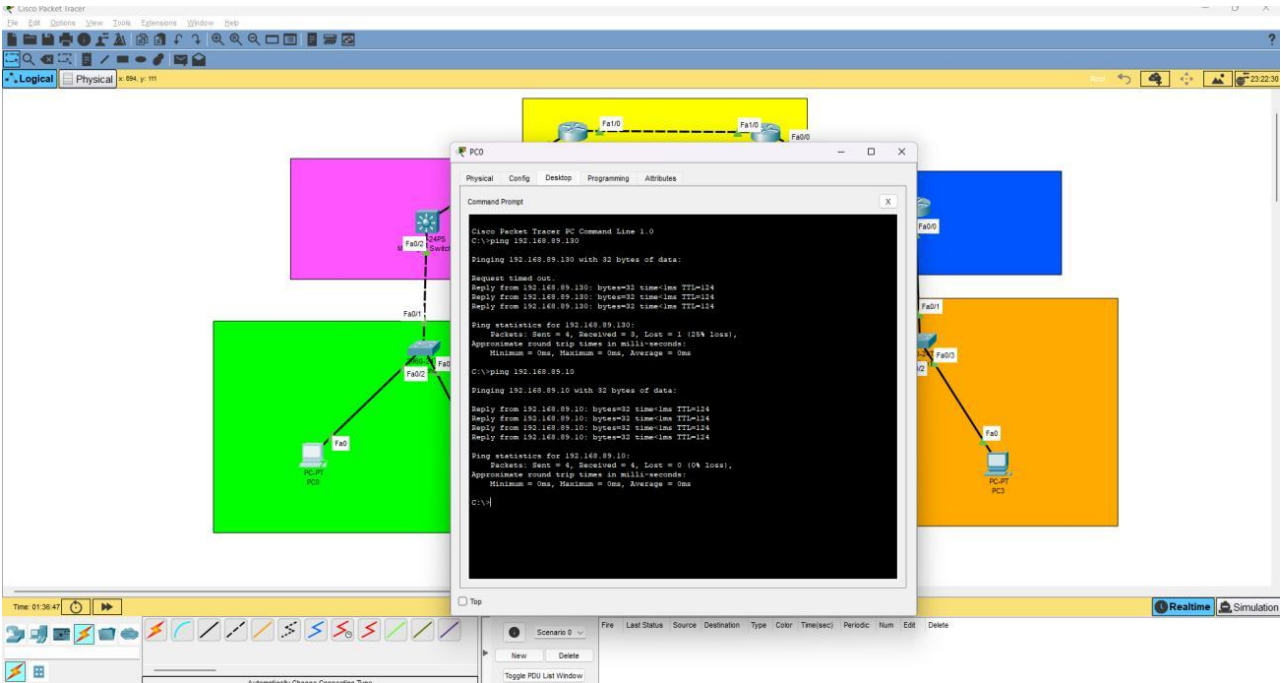
PC0 a PC3



PC1 a PC2



PC0 a PC2



Reflexión

¿Qué aprendí con esta tarea y cómo enfrenté los errores de configuración?

Aprendí mejor cómo se conectan las distintas partes de una red y cómo revisar si algo está mal. Cuando tuve errores fui probando poco a poco las interfaces, las VLANs, las rutas y los pings hasta encontrar qué configuración faltaba o estaba incorrecta.

¿Cuál es la conclusión técnica de la experiencia realizada?

Al final pude ver que para que toda la red funcione no basta con configurar una sola parte, sino que todo tiene que quedar conectado correctamente. También me sirvió para practicar los comandos de verificación y entender mejor cómo se comunican las diferentes redes.